

Product Requirements

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1. Overview

1.1 Purpose

Punching Power Pro is a fitness and boxing training app that utilizes the Apple Watch accelerometer to measure punch force and speed. It includes gamification elements, real-time analytics, and workout tracking to help users improve their striking power.

1.2 Target Audience

- Boxers and martial artists
- Fitness enthusiasts
- Athletes looking to measure and improve punch power

1.3 Key Features Summary

- Punch tracking (force, speed, velocity, frequency)
- Arcade and Workout modes
- Leaderboards and gamification
- Workout history and analytics
- Social sharing and third-party app integrations
- Apple Watch and mobile app sync

2. Functional Requirements

2.1 Onboarding and User Profile (P0)

Companion App Experience

Goal	Establish a profile and understand how to interact with the app
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Starting Point	The user will encounter this flow the first time they open the app. Or if they log out and want to create a new profile.
Steps/Actions	1. User opens Strike Force for the first time 2. User is presented with options to log in or create an account a. Account can be created via email and password b. Account creation can be done via OAuth using Apple iCloud account or Google accounts 3. If the user opts to sign in using email and password, present fields to enter this information. They do not need to confirm their email address. This is just used as an identifier linking a human to a profile. a. Ask for Age, DOB, Height, Weight 4. If the user signs in via OAuth, I think we should still ask for these things. But, I wonder if we also offer the option to use Healthkit and pull this data from their phone for more accurate and up-to-date data. It also simplifies the process and establishes the connection between Strikeforce and Apple Fitness so they can record workouts. Thinking of an interaction model very similar to Fitbod or Strava. 5. Optional: Upload Photo (can we automatically pull this using Gravatar, Apple, or Google as well? Give them the option to replace the photo if they don't like what was automatically pulled). 6. Show a screen including an animation and text describing what Arcade mode and Fitness mode are. 7. Goals: Understand punch force, punch speed, for use while working out, other (for use in product improvements and marketing efforts) 8. Brief TOS screen. We definitely want users to agree not to do crimes while using this app. a. I think we'll need to convey the hardware limitations of each model of Apple Watch and how much force can actually be measured.
Decision Points	This is a step-by-step workflow that blocks the user from engaging with the home view and core app functionality. Each step is a

	decision point to provide required information to or abandon. As they've already paid for the app, I think we won't see much drop-off.
Ending Point	The flow end upon agreement to the TOS (final step)
Annotations	• We want to request and require the minimum amount of data required to use the app. ○ We are requesting biometric data for use in calculating punch force and potentially as inputs to ML models we'll be training. ○ The photo can be used in the sharing functionality. I want to make that aspect a bit personal, "Chris has challenged you to a contest of strength..." will be more powerful with a face associated with it. • The tutorial screens are just to get people excited to use the app and understand early how they <i>should</i> use it. ○ We can/should encourage sharing/challenging others ○ We may want to briefly explain how the force is being measured (frequency of sampling per mode and what it actually means)

Watch App Experience

If the user opens the watch app before going through the onboarding flow in the companion app, display a message to the user, "Please open Strike Force on your iPhone to continue".

2.2 Home Screen and Core Application (P0)

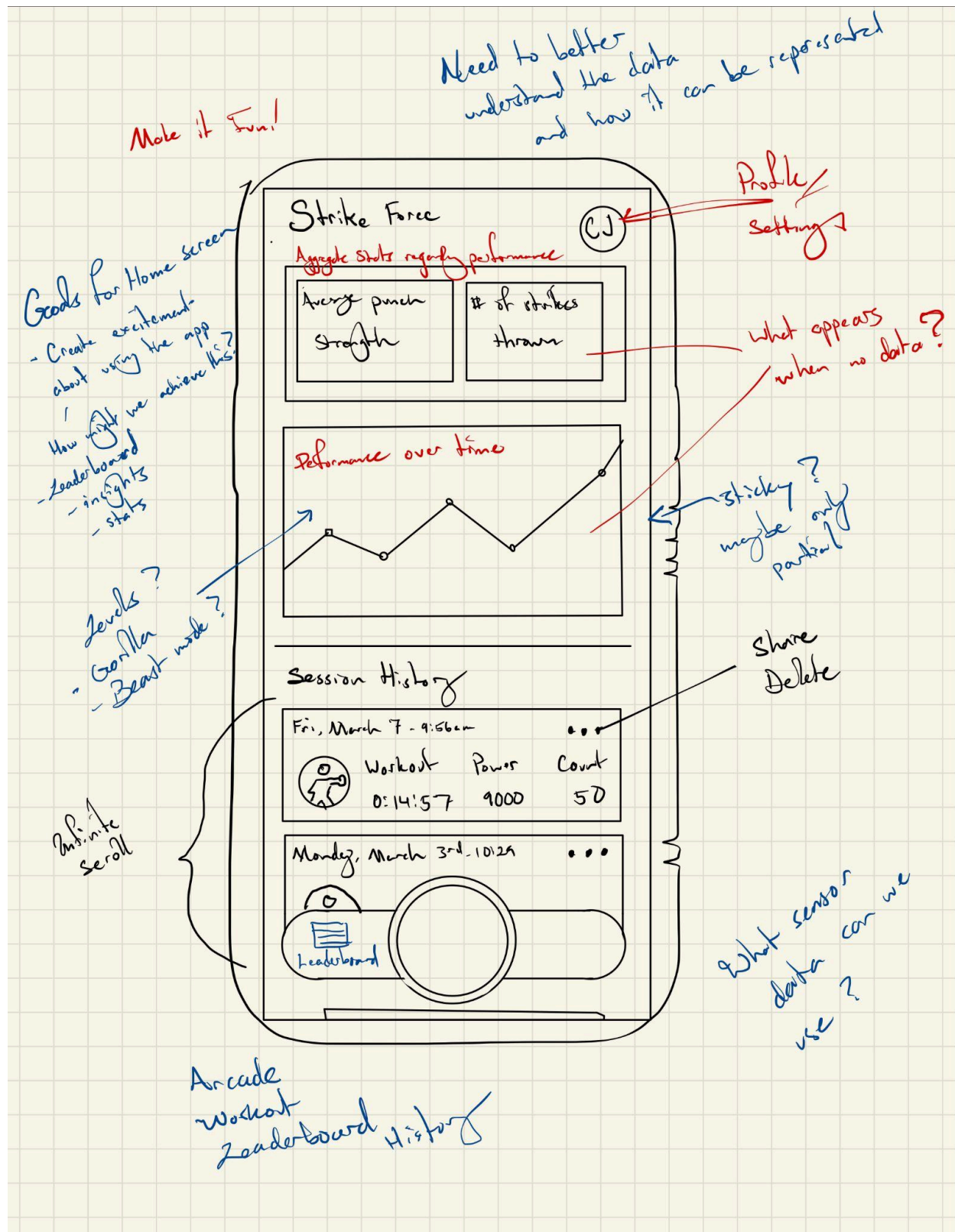
Companion App Experience

2.2.1 Home view

- The primary purpose of the companion app is:
 - To view / deep dive into session data
 - To share session data / sync to Apple Health, Post to Strava, etc.
 - To delete session data
 - To view the leaderboard
 - Update account/profile information
- User should be able to view statistics relevant to their engagement with Strike Force

- Personal Records
 - Arcade
 - Strongest Punch
 - Fastest Hands (most consecutive punches thrown in 30 seconds)
 - Challenges won
 - Fights won
 - I think we may want some engagement metrics here:
Session streak for instance

Rough concept sketch



Watch App Experience

- When the watch app is opened the user should be presented with cards containing the various workout modes. More details are listed below for each mode (data to display, animations, etc.).

2.3 Punch Tracking and Tracking Modes (P0)

- Measure force, speed, and frequency
- Display real-time data on Apple Watch and mobile app
- Confirm punch detection (minimum velocity threshold = 3G)

2.3.1 Arcade Mode

Watch Interface

- User can only start a session from the Watch interface
- **Single punch measurement (Power mode):** I think we may want helper text to help the user understand what to do.
 - Countdown timer (3,2,1, Go!) *Consider a haptic buzz*
 - Increase sampling rate to 100Hz (Whatever the device max is)
 - The High-Frequency Motion API that was introduced in 2023 (described here: <https://developer.apple.com/videos/play/wwdc2023/10179/>) needs an Apple Watch Ultra or Series 8 or higher to capture motion and accelerometer data at the higher frequency. There are rumors that the accelerometer can only capture up to 32G. Will want to verify. Documentation says otherwise.
 - Max G forces produced above 3G in a 5 second window.
 - We're going to capture x, y, & z and add these values to determine the total force of the punch. Sample Data: `Acceleration_Data`
 - The number in Arcade mode should animate:
 - Easing Function:
 - Uses an ease-out function (e.g., cubic, quartic, or exponential easing) to start fast and gradually slow down.
 - In CSS, JS, or SwiftUI, this is often `easeOut` or `easeOutExpo`.
 - Tweening Animation:
 - Interpolates between values over time.

- In Swift, this can be done using `UIView.animate` or SwiftUI's `.animation(.easeOut(duration: X))`.
- Rolling Counter Animation:
 - Mimics old-school mechanical counters where numbers rapidly roll and settle.
 - Often used in game score displays, slot machines, or arcade machines.
- After a punch is detected, display a fast-rising number that slows as it reaches the actual punch force.
- Add a slight "bounce" or flicker effect to make it feel dynamic.
- Combine with a flipping digit animation for an authentic arcade vibe.
- **Rapid consecutive punches (Speed mode):** I think we may want helper text to help the user understand what to do.
 - Countdown timer (3,2,1, Go!)
 - Begin timer counting down from 30
 - Display ongoing count
 - We're counting punches as the number of logs that go above 3G
 - TODO: Consider if we want to include any force stats here as well
- At the end of the session, blink three times and present options at the bottom of the watch screen (thinking buttons: Share, Challenge, Retry, Return to Menu)
 - **TODO:** Consider how this will integrate with the Leaderboard.
 - We could potentially include this in a post session summary screen.
 - Global ranking: ie. "You've thrown the 9th most powerful punch in the world"
 - Country ranking
 - Age bracket ranking
 - Gender? Do we want to bring this into the discussion?
 - I'm also thinking within these categories we can limit the time range. ie. "You've thrown the strongest punch in Germany today!"
 - Also... I want to include copy describing the force of a punch in funny terms. ie. *"Your strike (punch or strike? It is "Strike Force" after all...) has the same force as that of a kangaroo!"*

2.3.2 Workout Mode

- Start Workout button
- Countdown start (similar to Apple Fitness workouts and Arcade mode)
- Display core metrics on Apple Watch:
 - Timer (Elapsed Time)
 - punch count,
 - max force,
 - avg. force,
 - heart rate (optional)
- End workout button with option to cancel

2.3.3 Companion App Interface (similar behavior for both Arcade and Workout mode)

- If the app is open at the same time that a Strike Force session is initiated via Apple Watch, display a view with session statistics. Similar to YaoYao. Only we will want to keep this view open once the session is complete. If another session is initiated while the summary view of their previous session is still open, we will display statistics relevant to the new session and the previous one will be available in the session history section of the companion app.

2.4 History & Analytics (P0)

- Store past Arcade and Workout sessions
- Display personal records
- Visual data trends and performance insights
 - <https://www.scichart.com/ios-chart-features/>
 -  Punch Power Pro - Product Requirements Document ← Link to moodboard
 - Ideas/Inspiration:  Punch Power Pro - Product Requirements Document ← Competitors section
- Shareable workout summaries

2.5 Leaderboard & Challenges (P0)

2.5.1 Leaderboard

- Personal and global leaderboards
- Leaderboards exclusive to paid users

- Periodic global challenges (e.g., Boxing Day challenge: 100 punches in 1 min)

2.5.1 Challenge

- A challenge can be initiated following an arcade session.
- The user shares their performance with a friend or contact using a unique URL that allows them to respond to the challenge.
 - For a user without the app installed it would link them to the App Store to install
 - For a user with the app installed, it would link them to the challenge (tracked by challenge id possibly)
 - The challenge remains valid until completed. So, in the instance of the user without the app installed, they could click on the link again once the app is installed to respond to the challenge.
 - A challenge just directs the challenged to the respective arcade session (power or speed) allows them to complete it and then a winner is determined. A challenge is a type of session and will appear alongside other session data only with both user's result and a winner.
- A winner is determined by the greater force produced or punch count.

2.5.1 Fight!

- A fight can be initiated following a workout session in a similar way as a challenge as an option following session completion.
- A winner is determined by the sum total of force produced by punches in a workout session. So, one powerful punch could beat many less powerful punches for instance.

2.6 Social & Data Sharing (P0)

- Share performance metrics on social media (Strava, Instagram, TikTok, Reddit) using Open Graph Protocol (<https://ogp.me/>). We will want rich content included in what's shared to help drive growth.
- Export punch data and reports
- Integrate with Strava and Apple Health

2.7 Legal Compliance (P0)

- Liability disclaimers to mitigate risks
- Clear terms and conditions agreement

2.8 Account Settings (P0)

- Membership (if we support a premium tier)
- Getting Started Guide - the tutorial shown during onboarding
- Authorize Health Access
- Heart Rate Zone ← Maybe? If we support this view in the watch
- Siri Shortcut
- Watch Face
- Change App Icon
- Widget
- Apple Watch Ultra Action Button
- Backup & Restore (useful if we don't want to store all of this data in the cloud)
- Notifications (Let's say someone loses their rank on the leaderboard)
- General
- FAQs
- Share with friend (We'll want a deep link to the App Store)

2.9 Additional Features (Nice to haves)

Complications & Watch Faces (P2)

- Apple Watch complications for quick-launch (I think this can just be our app icon to open the app quickly. Will want to make sure)
- Custom watch faces displaying punch stats
- Siri shortcuts for voice-controlled actions

Video & Tutorials (P2)

- How-to guides for beginners
- Links to professional boxing training content (thinking Tony Jeffries as he has a video on how to punch harder and he has quite an audience. Definitely an opportunity here)

Affiliate Marketing (P2)

- Ability to enter promo-code for affiliate marketing (Would be another growth engine)
 - Probably only relevant if we support a premium tier though
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3. Non-Functional Requirements

3.1 Performance

- Low latency
- Offline mode - Let's say I'm in a cellular dead zone working out and I still want to record my session. We can display an "Offline" icon and certain features won't be available (leaderboard updates for instance) until a connection is restored.

3.2 Scalability

- Infrastructure prepared for global expansion. Millions of users (that's the goal, right?)

3.3 Security

- End-to-end encryption for user data
- GDPR and CCPA compliance
- Secure authentication (e.g., 2FA)

3.4 Usability

- Key features accessible within three taps
- Accessibility features (voice commands, colorblind-friendly UI)
- [Internationalization](#) (I think Apple may support some kind of out-of-the-box solution for this)
 - English
 - Spanish
 - Chinese
 - Japanese
 - German

3.5 Reliability

- 99.9% uptime
- Stable smartwatch-to-phone syncing

3.6 Compatibility

- iOS version support (How far back can we go while maintaining our feature set?)

- Apple Watch version support
- Regular updates to ensure OS compatibility

3.7 Maintainability

- Modular codebase for easy updates
- Developer documentation for onboarding

3.8 Legal & Ethical Standards

- Promote safe training practices
 - Warn users about overtraining risks
-

4. Dependencies

4.1 Internal Dependencies

- Backend services for data storage
- Machine learning models for punch analysis (possibly... I think there's a heuristic solution)

4.2 External Dependencies

- Apple Health & Strava APIs
 - Cloud infrastructure for leaderboard management
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5. Assumptions

- Apple Watch is the primary data collection device
 - Fitness tracking is done via Apple Health / Healthkit
 - Users will consent to data collection for analytics and sharing
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6. Risks & Mitigations

6.1 Data Security & Privacy

Risk: Non-compliance with GDPR/CCPA leading to legal issues **Mitigation:** Implement encryption, conduct security audits, consult privacy experts

6.2 Personal Injury & Liability

Risk: Users overexerting themselves and causing injuries **Mitigation:** Clear disclaimers, proper training tutorials, excessive force warnings

6.3 Scalability Issues

Risk: Rapid adoption leading to performance bottlenecks **Mitigation:** Cloud-based auto-scaling infrastructure, load testing, staged rollouts (I've heard of teams waking up to enormous AWS bills after their app went viral. Let's do some scenario planning and cost estimation here.)

6.4 Product Expectations vs. Reality

Risk: Users expecting unrealistic performance insights **Mitigation:** Clearly define feature scope, thorough QA testing, ongoing user feedback loop

7. Open Questions & TODOs

- ~~What additional sensors can we leverage?~~ Gyroscope for rotational force?
 - **Should we offer geolocation tracking?** Likely not necessary
 - **How do we rank leaderboard entries?** Raw force vs. force relative to weight?
 - **Determine gamification feedback phrasing** (e.g., "You punch as hard as a kangaroo!")
 - Validate price sensitivity with user testing
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8. Mockups & Wireframes

- To be added based on the finalized UX design